Configuration differences

Setting	Reference Base3a	Main port	Effect / caveat
Producer revision	e99efbb through120; resumed965a025	Main662507f9 + required imports; 489bd78dd5	Implementation/lineage differs; measure quality.
Trainer / rollout	Legacy DAPO / synchronous vLLM	V1 sync / AgentLoop + TransferQueue	Retain source ReMax objective within main design.
data.chunk_load_mode	sample	indexed	Requested branch setting; six representative real probes matched.
Historical training scorer	Legacy through120, current thereafter	Requested dffabc scorer from0	Uniform validation rescore does not undo historical rewards.

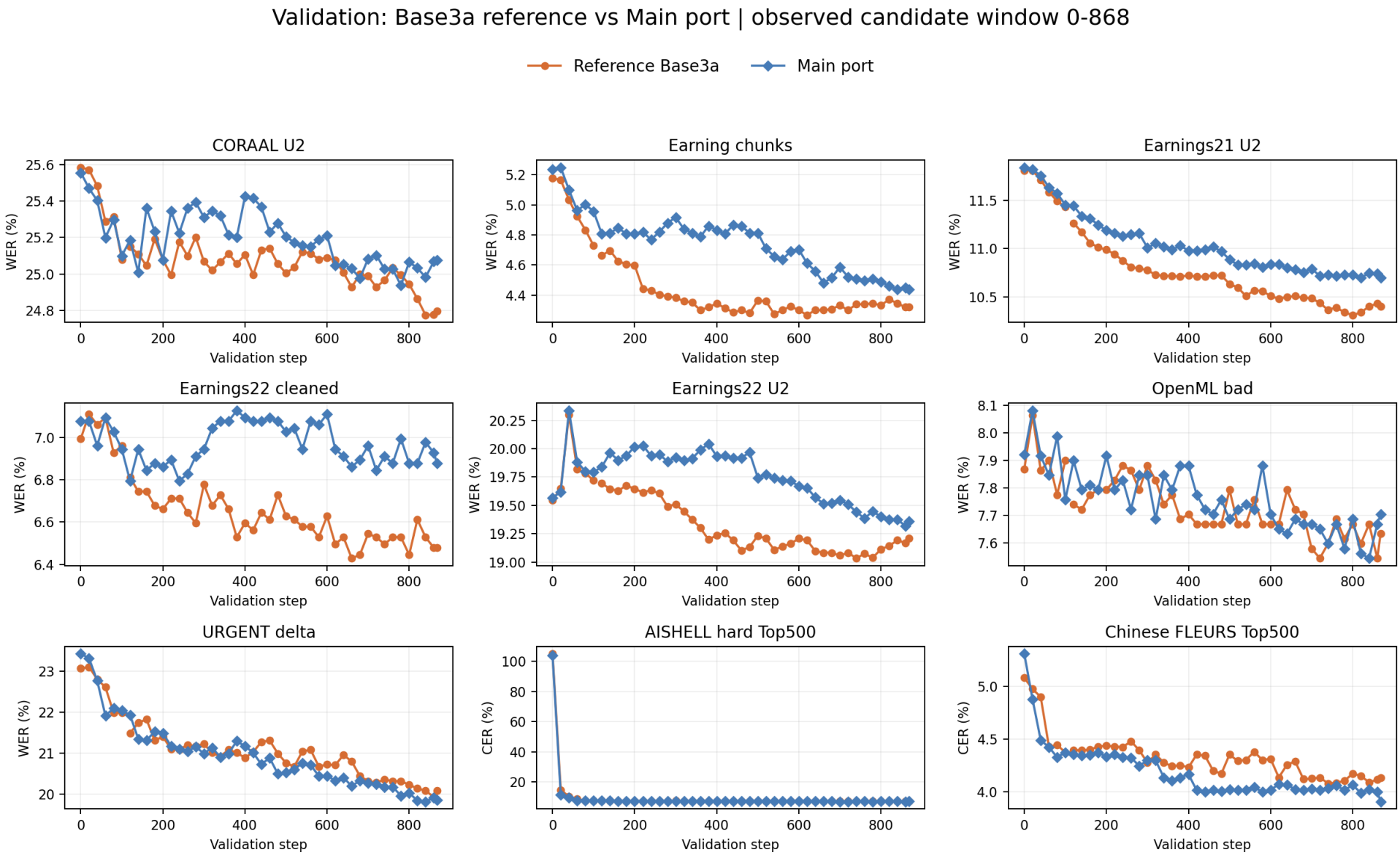
Multivariable implementation/lineage comparison. Shared controls and full configs are retained separately in evidence.

Submission base3a-main-training-20261113-a4; SUCCEEDED; optimizer 868/868. Training868, final validation/checkpoint/export868 and all three full35 evaluations complete. Validation868 and full35 regression gates are assessed separately; completed execution does not imply parity.

Final Validation868: 4/9 pass; full35@260: 31/35 pass; full35@868: 29/35 pass. Every dataset must meet the +0.1pp threshold. Historical scorer differences prevent a causal attribution of quality changes.

Shutdown warning retained: DataLoader worker-kill traceback in Python weakref exit finalizer after final868 metrics/validation and W&B sync; Ray driver exit0. Worker-kill cause is unresolved; independent checkpoint CPU/loader evidence is retained. No actual distributed checkpoint reload claimed.

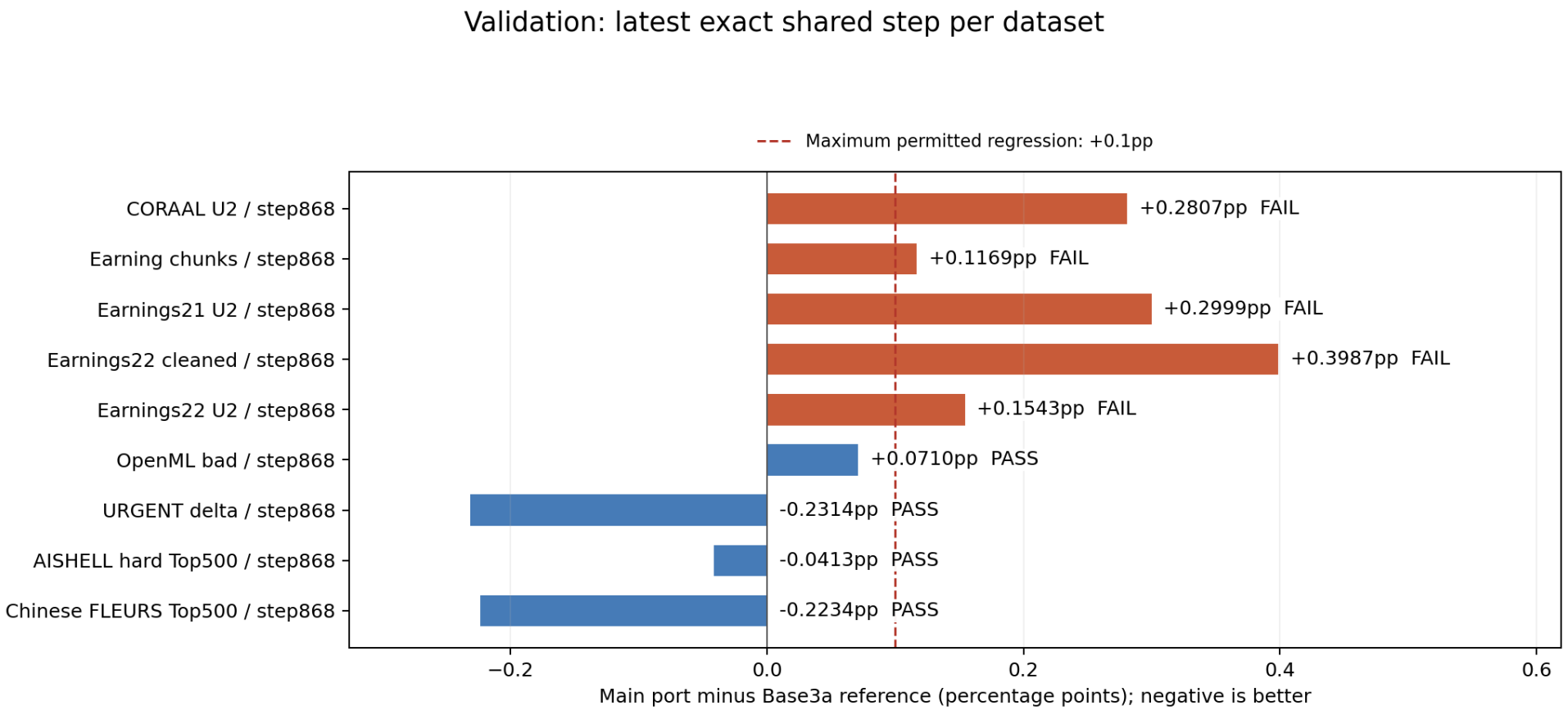
Validation



optimizer 868/868; latest completed validation 868; lower is better. Step0 precedes training; observed points only; no smoothing.

Candidate completed validation steps: [0, 20, 40, 60, 80, 100, 120, 140, 160, 180, 200, 220, 240, 260, 280, 300, 320, 340, 360, 380, 400, 420, 440, 460, 480, 500, 520, 540, 560, 580, 600, 620, 640, 660, 680, 700, 720, 740, 760, 780, 800, 820, 840, 860, 868]. Current window shown on common axes; lower is better. Reference is uniformly rescored. Step0 precedes training; prior retries remain archived separately.

Validation: per-dataset regression gate



Exact shared validation steps only; maximum permitted regression+0.1 percentage point per dataset. All prior retry results remain preserved; no best-retry selection.

Validation: latest exact shared step per dataset

Dataset / step	Reference %	Main %	Delta pp	Coverage / gate
coraal_u2 / 868	24.7967	25.0774	+0.2807	298 segments/298 groups; FAIL
earning_chunk_verb / 868	4.3196	4.4365	+0.1169	341 segments/6 groups; FAIL
earnings21_u2 / 868	10.4010	10.7009	+0.2999	355 segments/355 groups; FAIL
earnings22_cleaned_aa_u2 / 868	6.4784	6.8771	+0.3987	100 segments/100 groups; FAIL
earnings22_u2 / 868	19.2068	19.3611	+0.1543	332 segments/332 groups; FAIL
openml_bad / 868	7.6322	7.7032	+0.0710	393 segments/393 groups; PASS
urgent2024_delta / 868	20.0819	19.8505	-0.2314	283 segments/283 groups; PASS
zh_aishell1_top500_2611r2 / 868	7.1478	7.1065	-0.0413	500 segments/500 groups; PASS
zh_fleurs_top500_2611r2 / 868	4.1286	3.9051	-0.2234	500 segments/500 groups; PASS

Strict gate: at most +0.1 percentage point per dataset; matching raw cohort/references and fixed scorer required. Compound-aware normalization can legitimately change n_ref with hypotheses; both integer counts and equality flags remain in evidence.

Gate history: step0: 7/9 pass; step20: 8/9 pass; step40: 9/9 pass; step60: 9/9 pass; step80: 6/9 pass; step100: 8/9 pass; step120: 4/9 pass; step140: 4/9 pass; step160: 5/9 pass; step180: 4/9 pass; step200: 4/9 pass; step220: 4/9 pass; step240: 6/9 pass; step260: 4/9 pass; step280: 4/9 pass; step300: 4/9 pass; step320: 3/9 pass; step340: 3/9 pass; step360: 4/9 pass; step380: 2/9 pass; step400: 2/9 pass; step420: 3/9 pass; step440: 4/9 pass; step460: 5/9 pass; step480: 4/9 pass; step500: 4/9 pass; step520: 4/9 pass; step540: 5/9 pass; step560: 5/9 pass; step580: 3/9 pass; step600: 4/9 pass; step620: 5/9 pass; step640: 5/9 pass; step660: 4/9 pass; step680: 5/9 pass; step700: 5/9 pass; step720: 3/9 pass; step740: 5/9 pass; step760: 5/9 pass; step780: 5/9 pass; step800: 4/9 pass; step820: 5/9 pass; step840: 5/9 pass; step860: 3/9 pass; step868: 4/9 pass. Training868, final validation/checkpoint/export868 and all three full35 evaluations complete. Validation868 and full35 regression gates are assessed separately; completed execution does not imply parity.

Validation: full OpenASR at step260 (1/4)

Original vs Main @ 260: OpenEN

Header	Original Base3a@260	Main reproduce@260	WERR
OpenEN	A	B	A->B
ami_clean	6.6253%	6.6071%	0.27%
earnings22-cleaned-aa-chunked	4.7093%	4.9432%	-4.97%
gigaspeech_clean	7.3644%	7.3861%	-0.29%
librispeech-clean	1.1527%	1.1565%	-0.33%
librispeech-other	2.6135%	2.6267%	-0.51%
monsoon_en_in	3.3085%	3.2921%	0.49%
spgispeech_clean	2.1236%	2.1013%	1.05%
voxpopuli-cleaned-aa	1.4229%	1.4566%	-2.37%
urgent2024	9.1620%	9.1512%	0.12%
avg	4.2758%	4.3023%	-0.62%

Chinese CER; other datasets WER. WERR = 1 - Main / Original. Strict +0.1pp gate is evaluated separately.

Exact step260 endpoints; 35 datasets / 192883 original segments. Strict +0.1pp gate: 31/35 pass. WERR is relative change; negative WERR is regression. Matched raw cohorts and text scoring code verified. Normalized denominator differs on 7 datasets; compound-aware normalization may depend on the hypothesis. Rates use each verified corpus denominator.

Validation: full OpenASR at step260 (2/4)

Original vs Main @ 260: OpenML de/es/fr/hi

Header	Original Base3a@260	Main reproduce@260	WERR
OpenML	A	B	A->B
de_fleurs	1.9179%	1.9126%	0.28%
de_mcv	1.8508%	1.8550%	-0.23%
de avg	1.8844%	1.8838%	0.03%
es_fleurs	1.5772%	1.5815%	-0.27%
es_mcv	2.1761%	2.1876%	-0.53%
es_mls	2.6698%	2.6860%	-0.61%
es avg	2.1410%	2.1517%	-0.50%
fr_fleurs	2.4213%	2.4433%	-0.91%
fr_mcv	4.1953%	4.1960%	-0.02%
fr_mls	2.2366%	2.2541%	-0.78%
fr avg	2.9511%	2.9645%	-0.45%
monsoon_hi_in	8.4397%	8.5278%	-1.04%
hi avg	8.4397%	8.5278%	-1.04%

Chinese CER; other datasets WER. WERR = 1 - Main / Original. Strict +0.1pp gate is evaluated separately.

Exact step260 endpoints; 35 datasets / 192883 original segments. Strict +0.1pp gate: 31/35 pass. WERR is relative change; negative WERR is regression. Matched raw cohorts and text scoring code verified. Normalized denominator differs on 7 datasets; compound-aware normalization may depend on the hypothesis. Rates use each verified corpus denominator.

Validation: full OpenASR at step260 (3/4)

Original vs Main @ 260: OpenML it/nl/pt

Header	Original Base3a@260	Main reproduce@260	WERR
OpenML	A	B	A->B
it_fleurs	0.9186%	0.9515%	-3.58%
it_mcv	1.8047%	1.8103%	-0.31%
it_mls	4.1729%	4.0930%	1.91%
it avg	2.2988%	2.2850%	0.60%
nl_fleurs	2.7456%	2.7214%	0.88%
nl_mcv	1.6584%	1.6234%	2.11%
nl_mls	3.8056%	3.8174%	-0.31%
nl avg	2.7365%	2.7207%	0.58%
pt_fleurs	1.9879%	1.9879%	0.00%
pt_mls	4.7729%	4.7864%	-0.28%
pt avg	3.3804%	3.3872%	-0.20%
avg	3.4046%	3.4172%	-0.37%

Chinese CER; other datasets WER. WERR = 1 - Main / Original. Strict +0.1pp gate is evaluated separately.

Exact step260 endpoints; 35 datasets / 192883 original segments. Strict +0.1pp gate: 31/35 pass. WERR is relative change; negative WERR is regression. Matched raw cohorts and text scoring code verified. Normalized denominator differs on 7 datasets; compound-aware normalization may depend on the hypothesis. Rates use each verified corpus denominator.

Validation: full OpenASR at step260 (4/4)

Original vs Main @ 260: OpenLong + OpenExt

Header	Original Base3a@260	Main reproduce@260	WERR
OpenLong	A	B	A->B
earnings21	9.0927%	9.1931%	-1.10%
earnings22	9.6769%	9.8883%	-2.19%
coraal	12.9569%	13.0362%	-0.61%
avg	10.5755%	10.7059%	-1.23%
OpenExt	A	B	A->B
zh_fleurs	2.5922%	2.5603%	1.23%
zh_aishell1	1.4375%	1.4451%	-0.53%
zh avg	2.0148%	2.0027%	0.60%
vi_fleurs	2.7400%	2.7475%	-0.28%
vi_mcv	7.8940%	7.9136%	-0.25%
vi avg	5.3170%	5.3306%	-0.26%
avg	3.6659%	3.6666%	-0.02%

Chinese CER; other datasets WER. WERR = 1 - Main / Original. Strict +0.1pp gate is evaluated separately.

Exact step260 endpoints; 35 datasets / 192883 original segments. Strict +0.1pp gate: 31/35 pass. WERR is relative change; negative WERR is regression. Matched raw cohorts and text scoring code verified. Normalized denominator differs on 7 datasets; compound-aware normalization may depend on the hypothesis. Rates use each verified corpus denominator.

Validation: full OpenASR at step868 (1/4)

Original vs Main @ 868: OpenEN

Header	Original Base3a@868	Main reproduce@868	WERR
OpenEN	A	B	A->B
ami_clean	6.5767%	6.5500%	0.41%
earnings22-cleaned-aa-chunked	4.4300%	4.6379%	-4.69%
gigaspeech_clean	7.3128%	7.3309%	-0.25%
librispeech-clean	1.1433%	1.1546%	-0.99%
librispeech-other	2.6286%	2.6381%	-0.36%
monsoon_en_in	3.3023%	3.3003%	0.06%
spgispeech_clean	2.2224%	2.1681%	2.44%
voxpopuli-cleaned-aa	1.3610%	1.3441%	1.24%
urgent2024	8.9786%	8.8169%	1.80%
avg	4.2173%	4.2157%	0.04%

Chinese CER; other datasets WER. WERR = 1 - Main / Original. Strict +0.1pp gate is evaluated separately.

Exact step868 endpoints; 35 datasets / 192883 original segments. Strict +0.1pp gate: 29/35 pass. WERR is relative change; negative WERR is regression. Matched raw cohorts and text scoring code verified. Normalized denominator differs on 8 datasets; compound-aware normalization may depend on the hypothesis. Rates use each verified corpus denominator.

Validation: full OpenASR at step868 (2/4)

Original vs Main @ 868: OpenML de/es/fr/hi

Header	Original Base3a@868	Main reproduce@868	WERR
OpenML	A	B	A->B
de_fleurs	1.8642%	1.8856%	-1.15%
de_mcv	1.8815%	1.8923%	-0.57%
de avg	1.8728%	1.8889%	-0.86%
es_fleurs	1.5687%	1.5644%	0.27%
es_mcv	2.1730%	2.1845%	-0.53%
es_mls	2.7146%	2.7056%	0.33%
es avg	2.1521%	2.1515%	0.03%
fr_fleurs	2.4983%	2.4487%	1.99%
fr_mcv	4.2163%	4.2249%	-0.20%
fr_mls	2.2471%	2.2506%	-0.16%
fr avg	2.9872%	2.9747%	0.42%
monsoon_hi_in	8.1535%	8.2184%	-0.80%
hi avg	8.1535%	8.2184%	-0.80%

Chinese CER; other datasets WER. WERR = 1 - Main / Original. Strict +0.1pp gate is evaluated separately.

Exact step868 endpoints; 35 datasets / 192883 original segments. Strict +0.1pp gate: 29/35 pass. WERR is relative change; negative WERR is regression. Matched raw cohorts and text scoring code verified. Normalized denominator differs on 8 datasets; compound-aware normalization may depend on the hypothesis. Rates use each verified corpus denominator.

Validation: full OpenASR at step868 (3/4)

Original vs Main @ 868: OpenML it/nl/pt

Header	Original Base3a@868	Main reproduce@868	WERR
OpenML	A	B	A->B
it_fleurs	0.9138%	0.9468%	-3.61%
it_mcv	1.8092%	1.8182%	-0.49%
it_mls	4.1534%	4.2001%	-1.13%
it avg	2.2921%	2.3217%	-1.29%
nl_fleurs	2.6740%	2.6375%	1.36%
nl_mcv	1.6374%	1.5954%	2.56%
nl_mls	3.8168%	3.7557%	1.60%
nl avg	2.7094%	2.6629%	1.72%
pt_fleurs	1.9878%	1.9787%	0.46%
pt_mls	4.3932%	4.4881%	-2.16%
pt avg	3.1905%	3.2334%	-1.35%
avg	3.3368%	3.3502%	-0.40%

Chinese CER; other datasets WER. WERR = 1 - Main / Original. Strict +0.1pp gate is evaluated separately.

Exact step868 endpoints; 35 datasets / 192883 original segments. Strict +0.1pp gate: 29/35 pass. WERR is relative change; negative WERR is regression. Matched raw cohorts and text scoring code verified. Normalized denominator differs on 8 datasets; compound-aware normalization may depend on the hypothesis. Rates use each verified corpus denominator.

Validation: full OpenASR at step868 (4/4)

Original vs Main @ 868: OpenLong + OpenExt

Header	Original Base3a@868	Main reproduce@868	WERR
OpenLong	A	B	A->B
earnings21	8.5751%	8.8070%	-2.71%
earnings22	9.3638%	9.5828%	-2.34%
coraal	12.7100%	12.8845%	-1.37%
avg	10.2163%	10.4248%	-2.04%
OpenExt	A	B	A->B
zh_fleurs	2.5458%	2.4676%	3.07%
zh_aishell1	1.3907%	1.3783%	0.89%
zh avg	1.9683%	1.9230%	2.30%
vi_fleurs	2.7701%	2.8003%	-1.09%
vi_mcv	7.9627%	7.9431%	0.25%
vi avg	5.3664%	5.3717%	-0.10%
avg	3.6673%	3.6473%	0.55%

Chinese CER; other datasets WER. WERR = 1 - Main / Original. Strict +0.1pp gate is evaluated separately.

Exact step868 endpoints; 35 datasets / 192883 original segments. Strict +0.1pp gate: 29/35 pass. WERR is relative change; negative WERR is regression. Matched raw cohorts and text scoring code verified. Normalized denominator differs on 8 datasets; compound-aware normalization may depend on the hypothesis. Rates use each verified corpus denominator.

Validation: French supplements at step260 and step868

Step260: original Base3a versus Main reproduce

French supplement: excluded from canonical means; included in strict35 gate

Supplementary French	Original Base3a@260	Main reproduce@260	WERR
eslo_telephony_fr	24.1422%	23.7564%	1.60%
accueil_ubs_fr	26.8592%	27.3575%	-1.86%

Chinese CER; other datasets WER. WERR = 1 - Main / Original. Strict +0.1pp gate is evaluated separately.

Step868: original Base3a versus Main reproduce

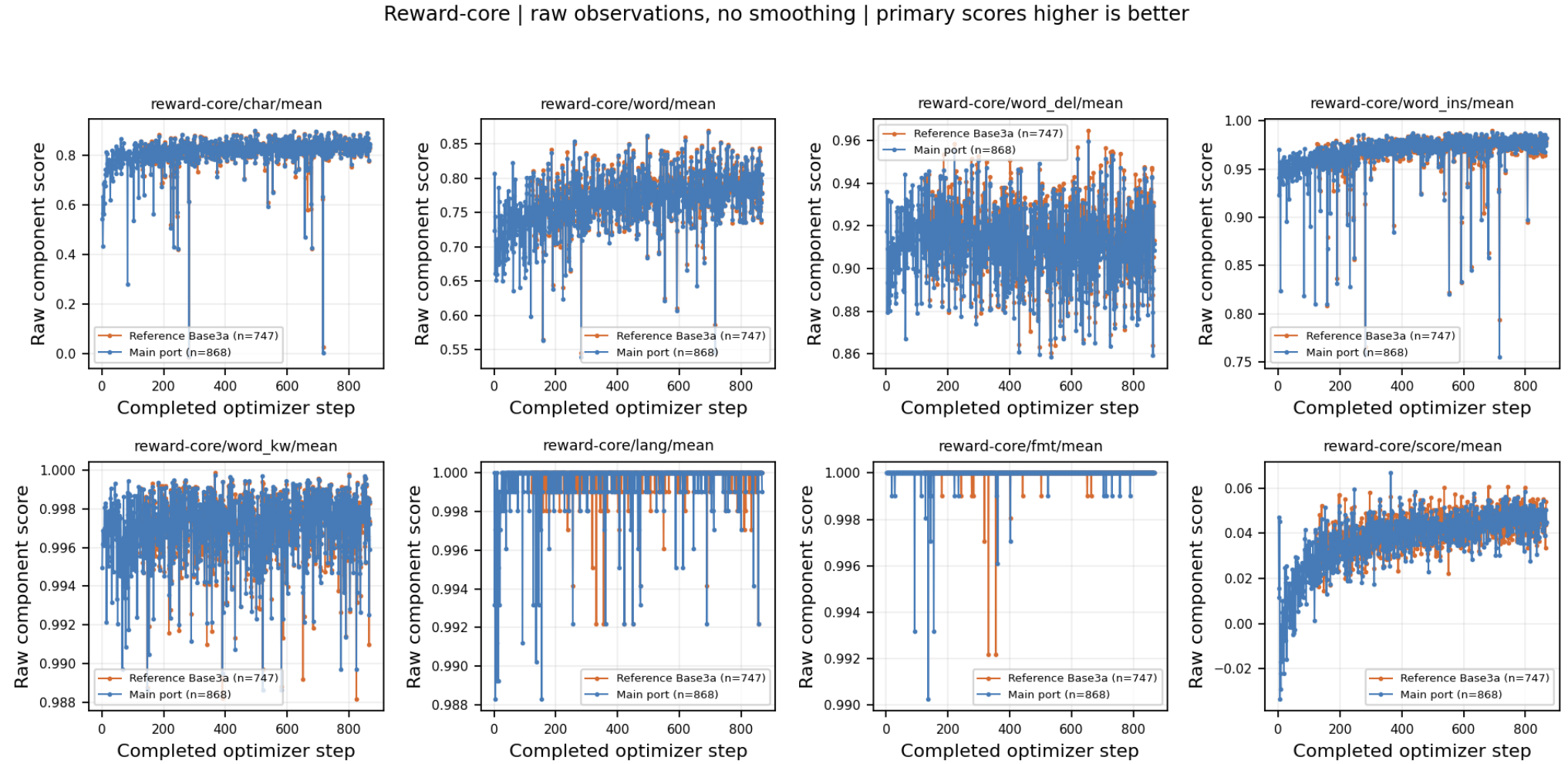
French supplement: excluded from canonical means; included in strict35 gate

Supplementary French	Original Base3a@868	Main reproduce@868	WERR
eslo_telephony_fr	24.1838%	25.1989%	-4.20%
accueil_ubs_fr	26.8072%	26.9732%	-0.62%

Chinese CER; other datasets WER. WERR = 1 - Main / Original. Strict +0.1pp gate is evaluated separately.

Both French datasets are excluded from canonical averages and retained in each strict35 gate.

Reward-core



Batch diagnostics, not validation. Reference legacy1..120 is excluded from numeric comparisons because its normalizer differs; complete original history, including discarded121/122, is retained.

Reward-core: exact shared optimizer window

Component	Window / n	Reference mean	Main mean	Delta
reward-core/char/mean	[121, 867] / 747	0.819584	0.819074	-0.000509765
reward-core/fmt/mean	[121, 867] / 747	0.999962	0.999948	-1.43804e-05
reward-core/lang/mean	[121, 867] / 747	0.999731	0.999691	-3.92194e-05
reward-core/score/mean	[121, 867] / 747	0.041278	0.0407076	-0.000570397
reward-core/word/mean	[121, 867] / 747	0.774587	0.772519	-0.002068
reward-core/word_del/mean	[121, 867] / 747	0.91439	0.911408	-0.00298197
reward-core/word_ins/mean	[121, 867] / 747	0.969052	0.969536	+0.000483136
reward-core/word_kw/mean	[121, 867] / 747	0.997141	0.997132	-8.96943e-06

Arithmetic raw means at exact common observed steps; no interpolation. Primary raw scores are higher-better, in native scorer units.